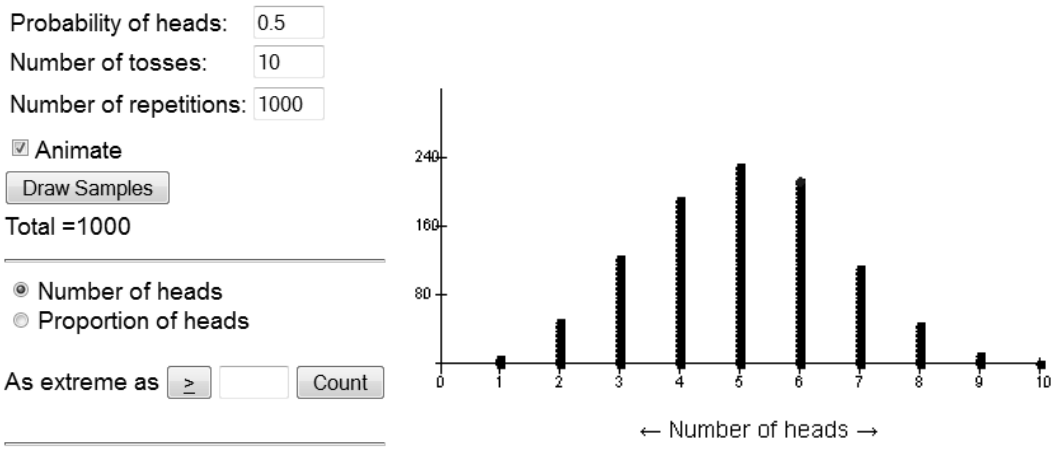
Solutions to Exploration 1.1

1. Can dogs understand human cues?
2. 10 trials
3. Whether Harley chose the correct cup; correct or incorrect; categorical
4. 10
5. 9/10; bar graph with one bar at 90% (correct) and another at 10% (incorrect)
6. Yes, 9/10 is greater than 50%.
7. Sure, it is possible.
8. Probably not; that's a fair bit larger than 50%.
9. 50%
10. More than 50%
11. Flip a coin 10 times to represent one set of Harley's 10 attempts. Record the number heads (heads = correct choice) out of 10 attempts
12. On average, we expect 5/10 times correctly choosing the cup if Harley is just guessing
13. (in-class part of the exploration)
14. 

    Yes, 9 heads rarely happened; 9 is in the tail of the distribution.
15. Yes, based on the simulated data in the dotplot, since 9/10 or more rarely occurs by chance, we can conclude that Harley would be unlikely to get 9/10 correct if he was just guessing.
16. Yes, the results appear to be statistically significant (unlikely to happen by chance).
17. No, the results suggest that Harley just guessing is not a plausible explanation for Harley's performance (9 out of 10 correct).

18.

Table 1.2: Parallels between real study and physical simulation

Coin flip	=	One of Harley's attempts
Heads	=	Harley picking the correct cup
Tails	=	Harley picking the incorrect cup
Chance of Heads	= 1/2 =	Long-run proportion of Harley picking the correct cup if he is just guessing
One repetition	=	one set of 10 simulated attempts by Harley

18. 9/10
19. We flipped a coin _10___ times and kept track of how many times it came up heads. We then repeated this process __999___ more times, each time keeping track of how many heads were obtained in each of the __10____ flips.
20. Because we rarely obtained a value of __9_____ heads when flipping the coin ___10___ times, this means that it is _____ (believable/unlikely) that Harley is just guessing, since if Harley was just guessing he _____ (rarely/often) would get a value like _9_____ correct out of _10_____ attempts.
21. Seems like Harley may be able to understand this experimenter when he bows; not necessarily all human cues, though.
22. a. 6/10 is not very unlikely b. No* it's not very unlikely that Harley gets 6/10 if he is just guessing. 6/10 happens frequently in the applet's dotplot. c. No, the result is not statistically significant. d. It plausible that Harley is just guessing which cup to pick in this case e. No, it just doesn't disprove it. Harley not understanding the mechanical arm (just guessing) is one plausible explanation for what we observed in the study.
23. 9/10 is more unusual than 6/10 because 5/10 is what happens "on average"--farther away from the typical outcome is less likely. Since the cup pointed to in each trial was randomly chosen and each of the two studies was conducted the same with the exception of who/what did the pointing, there is evidence the difference in the proportions correct between the studies was caused by the difference in human vs. mechanical pointing.
24. These two analyses suggest that Harley seems to understand human cues better than non-human cues. Further studies in a variety of situations with Harley (different kinds of cues, different nonhuman cues, etc.) and with other dogs are now needed to pursue this research area further.

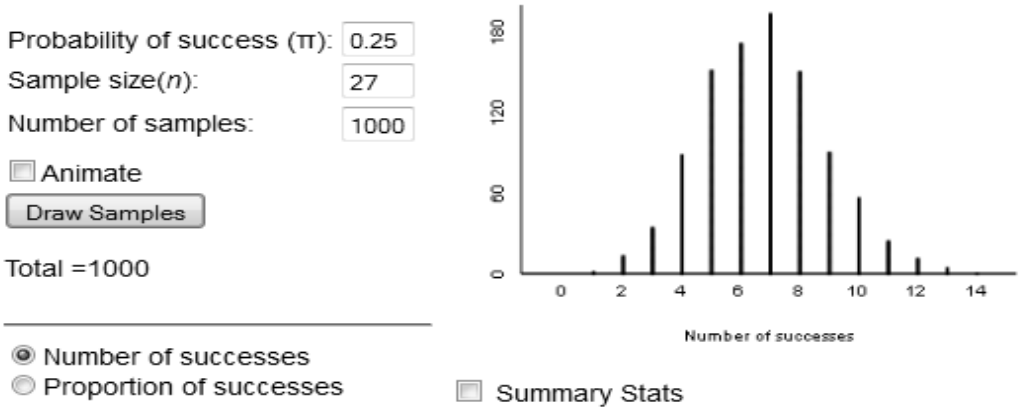
Solutions to Exploration 1.2

1. Do people really prefer tap water less than bottled water?
2. Each subject (person)
3. The variable is whether the person chose tap water as their favorite, or not. It is categorical.
4. Two
5. Parameter = long-run proportion of a subject in the study choosing tap water as their favorite when given a choice between 4 cups.
6. ¼
7. That it's less than ¼
8. Null hypothesis: the long-run proportion of a subject choosing tap water as their favorite is ¼ (Q6)

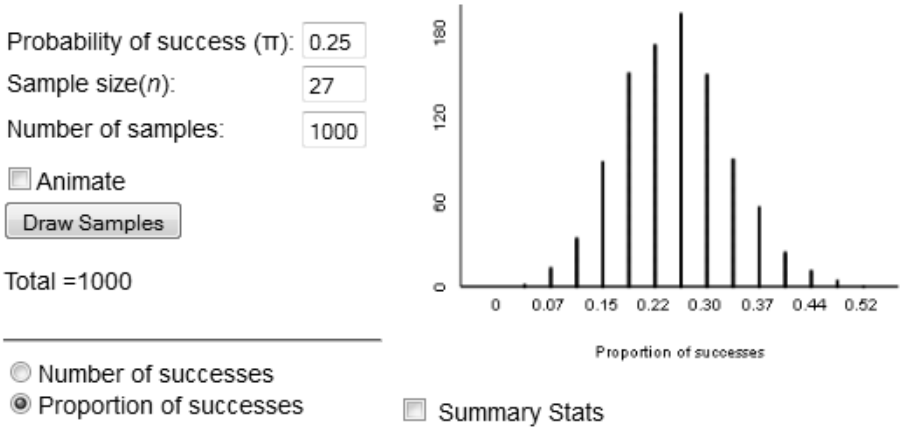
   Alt hypothesis: the long-run proportion of a subject choosing tap water as their favorite is < 1/4 (Q7)
9. 3/27
10. 0.111 (or 3/27)
11. 27
12. H_0: $\pi = ¼$

    H_a: $\pi<1/4$
13. Yes, $3/27 = 0.11 < 0.25$
14. Yes, it is possible.
15. Can't use a coin because the null hypothesis (by chance) probability is not ½
16. Shuffle the four cards. Randomly choose one. If black "success"; if not, "not success." Keep track of how many times a black card is chosen when you randomly choose a card 27 times. Repeat this process 1,000 times.
17. Dark (blue) region represents "success" (chose tap water); light (pink) region represents "failure" (chose bottled water). Spin the spinner 27 times

18.



Center = 6 or 7 (one quarter of 27), Variability = Most outcomes occur between 2 and 12 shape = bell-shaped curve



Same screen shot, only with proportion of successes checked. Center about 0.25, variability: most proportions between 0.07 and 0.44. Shape: bell-shaped

19. The outcome is near the tail, it is "kind of" in the tail of the distribution
20. Approximately 74/1000 (0.074) of the simulated sample proportions are 0.111 or less.
21. The p-value (0.074) gives us moderate evidence that people prefer tap water less than bottled (equivalently: moderate evidence against the null hypothesis)
22. The alternative hypothesis was less than.
23. While we have moderate evidence against the null, it is not strong so I would not say it is statistically significant.
24. If it were a significant result, I wouldn't be too comfortable generalizing this result to that many more people because we don't know much about who these 27 people are and how they might be different from a larger population of interest. We also don't know much about the water sources.

25. Further studies are needed with a more representative set of individuals. Could use tap water from different areas of the country to see if some tap waters were more easily distinguished from bottled than others.

25. a. more, b. Null hypothesis: $\pi = ¾$ Using words: Null: The long-run proportion of times bottled water is chosen is = ¾, c. alt hypothesis: $\pi >$ ¾ Using words: The long-run proportion of times bottled water is chosen is > ¾, d. 24/27=0.889, e. Prob = 0.75, sample size = 27, number of samples=1000, f. center = 13 or 14, variability = most outcomes are between 2 and 12, bell-shaped--pretty much the same as before, g. 0.889 or LARGER and then divide by 1000. h. p-value approximately 0.07, I = Probability of obtaining 0.889 or larger assuming null hypothesis is true, j=Moderate evidence against null hypothesis,

26. Yes, same result. Doesn't matter whether you focus on choose tap or choose bottled. Evidence level is still the same.

Solutions to Exploration 1.3

1. Each student in the class
2. Variable: Tim on left? Yes/no, categorical variable.
3. Null: The long-run proportion that a student in your class assigns the same (Tim) name to the picture on the left is 50%.

   Alt: The long-run proportion that a student in your class assigns the same (Tim) name to the picture on the left is more than 50%.

   Null: $\pi = 0.50$

   Alt: $\pi > 0.50$

   π = The long-run proportion that a student in your class would assign the same name to the face on the left.
4. For the solutions, we will use a class that had 18 students out of 25 total students put Tim on left. Your class may be different, changing answers to subsequent questions.
5. Number (Tim on left): 18, Sample size =25, proportion = 18/25 = 0.72
6. a. Probability=0.50, b. Sample size=25, c. Number of samples=1000
7. a. 0.72 or larger b. 0.018 c. We have strong evidence that the long-run proportion that a student in your class assigns the name "Tim" to the name of the picture on the left is more than 50%.
8. a. Close to 50%. B. 50%, because that is the null hypothesized value of the long-run proportion. c. For my set of 1,000 simulations, SD = 0.099, d. 0.72, e. (0.72-0.5)/0.099 = 2.22
9. a. Strong evidence b. Similar strength of evidence as obtained using the p-value
10. When presented two pictures and two names, the majority of students in our class (18/25) assigned the name Tim to the picture on the left, a large enough proportion that it provides strong evidence that the long-run proportion of students in our class assigning the name Tim to the picture on the left is above 50%. Numerous next steps exist in this research area: (a) Trying to elucidate which facial features are associated with which names and why (test more face/name pairs); (b) Doing the study with other people to see if the results hold up in different populations (different SES/ethnicities).
11. a. smaller, b. larger, c. stronger
12. a. It will be negative since the value is less than the mean, and so the numerator of the standardized statistic formula will be negative (the denominator will always be positive because standard deviation is always positive), b. There is no evidence against the null in favor of the alternative because the alternative says "greater than" and the observed data is not greater than.

Solutions to Exploration 1.4

1. Null: The probability of the player wearing the red uniform winning is 50%.
   Alt: The probability of the player wearing the red uniform winning is greater than 50%.
2. Null: $\pi = 50\%$, Alt: $\pi > 50\%$
3. a. 248/457 = 54.3%, b. Let a coin flip represent a match, heads = red wins, tails = blue wins; flip a coin 457 times keeping track of the number of heads; repeat 1,000 times recording the number of heads in each set of 457 flips. c. p-value = 0.03, mean = 0.499, SD = 0.024
4. Null: $\pi = 50\%$, Alt: $\pi \neq 50\%$ where π = the probability of the player wearing the red uniform winning.
5. a. Both tails are shaded, not just one. b. p-value = 0.075 (about twice as large as before), c. 0.543-0.50 = 0.043, 0.50-0.043 = 0.457. Thus, 0.457 is the comparable value.
   d. 0.075/0.543/0.457/null hypothesis., e. Switching from a one-sided to two-sided test increases the p-value (by approximately twice as much) and, thus, decreases the evidence against the null hypothesis.
6. a. If we use 0.57 as the statistic the p-value is 0.003. b. Smaller, because the statistic (57%) moved farther away from the null hypothesized value (50%) than before (54.3%) c. The farther the statistic is from the null hypothesized value, the smaller the p-value will be (strength of evidence increases)
7. a. Center = 0.50 (similar to before), variability (SD = 0.03) is larger b. p-value=0.058, larger than the original p-value because there is more variability in the null distribution c. All things being equal as the sample size increases, the strength of evidence increases (p-value decreases).

Solutions to Exploration 1.5

1. Your choice of heads or tails
2. a. Each student in the class, b. whether the students picks heads or tails, c. parameter π = long-run proportion of times students in our class will pick heads, d. 50%, null hypothesis e. larger than 50%, alternative hypothesis
3. (varies from class to class)♀ we will use 16 out 25 = 64%
4. Sample size = 83 + 25 = 108, sample proportion = 70/108 = 64.8%
5. A. Shape = bell-shaped, yes, this is a familiar shape, center/mean = 0.501, that makes sense because that is the null hypothesized value, mean = 0.501, SD = 0.048 b. p-value = 0.001 provides strong evidence against the null hypothesis c. z = 3.08, p-value = 0.001, Yes, similar to what we got from p-value
6. Yes, the blue-shaded region seems to match the dotplot from the simulation fairly well.
7. Yes, the sample size condition is met. There are 70 (> 10) people who picked heads and 38 (> 10) people who picked tails.
8. $\sqrt{\frac{0.5(1-0.5)}{108}}$. = 0.048. The theoretical SD (0.048) is very close to the simulated SD (see question 5a, simulated SD = 0.048)
9. (0.648-0.5)/0.048 = 3.08. This is very close to what we got when simulating.
10. The p-value from theory (normal approximation) is 0.0010; we got 0.001 when simulating. They are very similar.
11. Because the validity condition is met (sample size is large) these values are all similar. The values would be different if the sample size was small (validity condition not met).
12. Null: People pick heads 70% of the time; Alt = People don't pick heads 70% of the time; z =1.18, p-value = 0.2396. We don't have evidence that people don't pick heads 70% of the time; the null hypothesis (people pick heads 70% of the time) is plausible.
13. Two sided p-value from simulation is approximately 0.07.
14. The validity conditions aren't met. There are only 7 successes (people who picked heads) and one failure (person who didn't pick heads). Neither value is more than 10, and we need both to be larger than 10.
15. The p-value from the normal approximation method is 0.0339, different than the 0.07 from simulation.